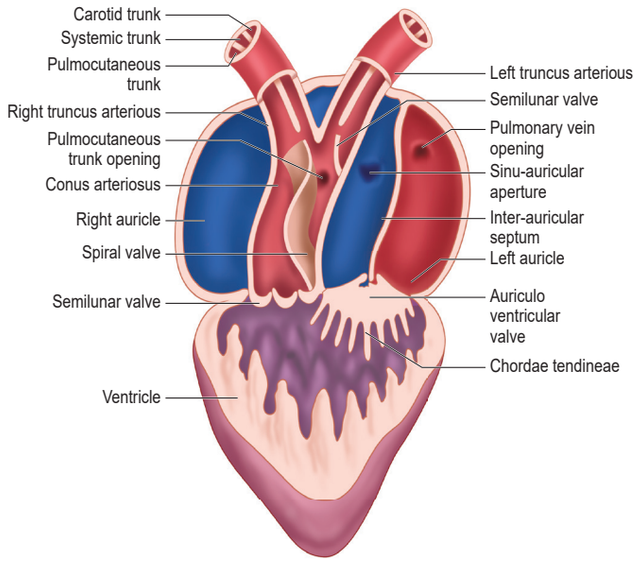


Figure 4.20 *Rana hexadactyla - Internal Structure of Heart*

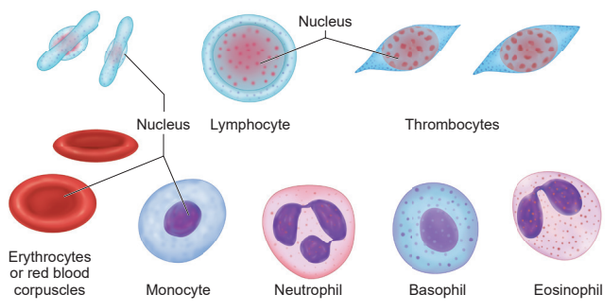


Figure 4.21 *Rana hexadactyla* – Blood cells

The **blood** consists of **plasma** [60%] and blood **cells** [40 %] includes red blood cells, white blood cells, and platelets. RBCs are loaded with red pigment, nucleated and oval in shape. Leucocytes are nucleated, and circular in shape (Figure 4.21).

The Nervous System

The Nervous system is divided into the Central Nervous System [CNS], the Peripheral Nervous System [PNS] and the Autonomous Nervous System [ANS]. **Peripheral Nervous System** consists of 10 pairs of **cranial nerves** and 10 pairs of **spinal nerves**. **Autonomic Nervous System** is divided into **sympathetic** and **parasympathetic** nervous system. They control involuntary functions of **visceral organs**. CNS consists of the Brain and Spinal cord. Brain is situated in the cranial cavity and covered by two meninges called piamater and duramater. The brain is divided into forebrain, midbrain and hindbrain. Fore brain (Prosencephalon) is the anterior most and largest part consisting of a pair of **olfactory lobes** and **cerebral hemisphere** (as Telencephalon) and a **diencephalon**. Anterior part of the olfactory lobes is narrow and free but is fused posteriorly. The **olfactory lobes** contain a small cavity called **olfactory ventricle**. The mid brain (Mesencephalon) includes two large, oval **optic lobes** and has cavities called **optic ventricles**. The hind brain (Rhombencephalon) consists of the **cerebellum** and **medulla oblongata**. Cerebellum is a narrow, thin transverse band followed by **medulla oblongata.** The medulla oblongata passes out through the **foramen magnum** and continues as **spinal cord,** which is enclosed in the vertebral column (Figure 4.22).

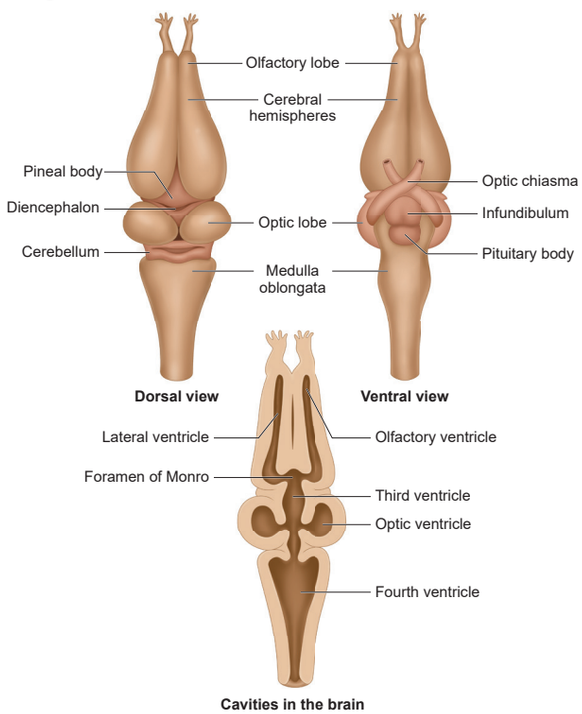


Figure: 4.22 *Rana hexadactyla* – Brain dorsal and ventral view

TN_GOVT_XI_Zoology_Ch04.indd 67 09-01-2020 5.14.29 PM